

Mbalo Platform: Detailed Architecture & Actor Overview

1. Introduction to Mbalo

Mbalo is a premium, interactive, gamified mathematics learning platform designed for Grades 1 to 7.

It uses a Safari theme and a friendly mascot to create an engaging learning experience.

The platform emphasizes active recall, minimal text, and strong visual/audio feedback.

2. Platform Actors

A. The Learner (Primary Actor)

- Selects grade and sets up profile
- Navigates course map and modules
- Completes quizzes and lessons
- Uses practice hub tools
- Tracks progress and achievements

B. The Parent / Guardian (Secondary Actor)

- Views progress dashboard
- Reviews learner struggles
- Manages account and settings

C. Administrator / Educator (Backend Actor)

- Manages curriculum data
- Oversees learner analytics

3. Core Features & Functionality

3.1 Authentication & Profile Management

- Email/password and Google login
- Grade-based curriculum routing
- Profile customization and dark mode

3.2 Dynamic Curriculum Engine

- Structured hierarchy (Grade → Chapter → Module → Lesson)
- Visual course map with progression tracking

3.3 Interactive Quiz Engine

- Multiple question types (MCQ, emoji counting, drag-and-drop, etc.)
- Number lines and column math tools
- Text-to-speech and calculator support

3.4 Practice Hub & Mixers

- Randomized math practice tools
- Visual explanations for answers
- Column math simulations
- Multiplication learning phases

3.5 Adaptive Analytics

- Tracks repeated mistakes ('struggles')
- Awards milestones and ranks
- Displays mastery progress bars

4. Technical Architecture

Frontend: React, TypeScript, Vite

UI/Styling: CSS, Framer Motion, SweetAlert2, Lucide React

Backend: Firebase (Auth + Firestore)

Testing: Vitest, React Testing Library